

Nayan Kumar

+91-8709551101 nayanparashar1@gmail.com linkedin.com/in/nayan-parashar github.com/Coder-nayan07

Education

Indian Institute of Technology, Dhanbad

Integrated M.Tech in Mathematics and Computing (CGPA: 8.54/10.00)

May 2028

Dhanbad, Jharkhand

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Probability & Statistics, Linear Algebra, GPU Computing.

Publications

GRIM (Grasp Re-alignment via Iterative Matching)

Co-author

AAAI 2026 (Oral)

Project Page | arXiv

Experience

Center for Intelligent Robotics - IIIT Allahabad

Research Intern

Mar 2025 – Sep 2025

Prayagraj, Uttar Pradesh

- Developed a multi-view RGB-D perception pipeline and constructed 3D feature fields using back-projected DINOv2 embeddings.
- Benchmarked 3 state-of-the-art robotic manipulation baselines against our novel approach, proving a 5–8% absolute increase in adversarial grasp success rates.
- Validated the system on a Kinova Gen3 Lite manipulator through 20+ real-world grasping trials.

Projects

AI Interview Platform | Python, LangChain, Groq API, Pydantic, PyMuPDF, Whisper | GitHub

- Developed a candidate profiling pipeline that transformed resumes and job descriptions into structured profiles in <1.2 s using PyMuPDF and Pydantic.
- Designed a 5-stage interview orchestration engine for interview planning, adaptive questioning, answer evaluation, and automated report generation.
- Engineered a low-latency multimodal interface combining asynchronous STT/TTS models with a finite-state machine, delivering sub-second conversational turnarounds.

Amazon ML Challenge 2025: Smart Product Pricing | Python, LightGBM, XGBoost, TF-IDF | GitHub

- Implemented a multimodal regression pipeline combining structured metadata and product descriptions, achieving a baseline SMAPE of 50.05%.
- Engineered 20k-dimensional TF-IDF features and performed feature ablation, reducing SMAPE to 46.03% and securing a Top 2% national ranking.

Technical Skills

Programming: Python, C++, C

Core CS: Data Structures & Algorithms, Object-Oriented Programming

Libraries: PyTorch, OpenCV, Open3D, NumPy, Pandas, LangChain, Pydantic, C++ STL

Developer Tools: Git, Linux, VS Code

Areas: Machine Learning, Computer Vision, Large Language Models (LLMs)

Achievements

- Competitive Programming – Codeforces Specialist (1560) | CodeChef 3-Star (1700)
- Solved 800+ algorithmic problems across Codeforces, LeetCode and GeeksforGeeks.
- Top 2% National Rank – Amazon ML Challenge 2025
- Top 3.3% Nationwide – CodeFest’25 (All-India ICPC-style Contest)

Present

Present

Oct 2025

Mar 2025

Leadership & Engagement

Senior Member: RoboISM (Robotics & AI Club, IIT Dhanbad).

Team Member: IIT (ISM) Dhanbad, Inter IIT Tech Meet 14 – Eternal (Autonomous Buggy Challenge).